Supplemental Document

Optics EXPRESS

Regulation-aware freeform headlamp reflector design with differentiable ray tracing: supplement

XUCHEN WEI,[1,] YUCHI HUO,[1] PENGFEI SHEN,[2] YIFAN PENG,[2,3] HUJUN BAO,[1] AND RUI WANG[1,4]

[1] *The State Key Lab of CAD&CG, Zhejiang University, Hangzhou 310058, China*
[2] *The University of Hong Kong, Hong Kong SAR, China*
[3] *evanpeng@hku.hk*
[4] *ruiwang@zju.edu.cn*

Supplement DOI: https://doi.org/10.6084/m9.figshare.29441531

Parent Article DOI: https://doi.org/10.1364/OE.566951

Regulation-aware freeform headlamp reflector design with differentiable ray tracing: supplemental document

This supplemental document contains six sections. In Section 1, we present the regulatory test reports for the four beam patterns generated in Section 3.1 of the main text. These reports detail the intensity values at the test points and test zones, and all four results successfully passed the regulatory tests, validating the effectiveness of our neural network method. In Section 2, we outline the configuration method for setting up the reference model in the LightTools software for the example reflector described in Section 3.2 of the main text. In Section 3, we provide the optical scene configuration and simulation results in LightTools for the driving-beam reflectors in Section 3.3 of the main text. In Section 4, we present the optical scene configuration and simulation results in LightTools for the passing-beam reflectors in Section 3.4 of the main text. In Section 5, we present the regulation test result tables of the 4 designed reflectors. Finally, in Section 6, we design an experimental B-Spline reflector to present the simulation deviation of outgoing rays between the B-spline surface and triangle mesh.

1. REGULATION TEST REPORTS OF NEURAL PROCESS RESULTS

Table S1. Test report for ECE R112 class-A driving-beam distribution

Test Item	(u, v)	Minimum(cd)	Maximum(cd)	Actual(cd)	Status
Point H-5L	(-5.0, 0.0)	3400.0	-	17545.6	PASSED
Point H-2,5L	(-2.5, 0.0)	13500.0	-	65123.1	PASSED
Point H-2,5R	(2.5, 0.0)	13500.0	-	62122.0	PASSED
Point H-5R	(5.0, 0.0)	3400.0	-	16702.7	PASSED
Imax	(0.1, -0.0)	27000	215000	212263.3	PASSED

Table S2. Test report for ECE R112 class-B driving-beam distribution

Test Item	(u, v)	Minimum(cd)	Maximum(cd)	Actual(cd)	Status
Point H-5L	(-5.0, 0.0)	5100.0	-	108743.1	PASSED
Point H-2,5L	(-2.5, 0.0)	20300.0	-	164682.7	PASSED
Point H-2,5R	(2.5, 0.0)	20300.0	-	165574.0	PASSED
Point H-5R	(5.0, 0.0)	5100.0	-	111868.4	PASSED
Imax	(0.1, -0.0)	40500	215000	212247.6	PASSED

Table S3. **Test report for ECE R112 class-A passing-beam distribution**

Test Item	(u, v)	Minimum(cd)	Maximum(cd)	Actual(cd)	Status
Point B 50 L	(-3.4, 0.6)	-	350.0	70.6	PASSED
Point BR	(-2.5, 1.0)	-	1750.0	102.7	PASSED
Point 75 R	(1.1, -0.6)	5100.0	-	7084.3	PASSED
Point 75 L	(-3.4, -0.6)	-	10600.0	1477.8	PASSED
Point 50 L	(-3.4, -0.9)	-	13200.0	2518.0	PASSED
Point 50 R	(1.7, -0.9)	-	5100.0	3907.7	PASSED
Point 50 V	(0.0, -0.9)	5100.0	-	8812.6	PASSED
Point 25 L	(-9.0, -1.7)	1250.0	-	4035.3	PASSED
Point 25 R	(9.0, -1.7)	1250.0	-	4575.4	PASSED
Point 7	(-8.0, 0.0)	65.0	-	396.0	PASSED
Point 8	(-4.0, 0.0)	125.0	-	254.5	PASSED
Group 1 - Point 1	(-8.0, 4.0)	-	-	550.4	-
Group 1 - Point 2	(0.0, 4.0)	-	-	372.9	-
Group 1 - Point 3	(8.0, 4.0)	-	-	12.2	-
Group 1 - Sum	-	190.0	-	935.5	PASSED
Group 2 - Point 4	(-4.0, 2.0)	-	-	266.6	-
Group 2 - Point 5	(0.0, 2.0)	-	-	203.5	-
Group 2 - Point 6	(4.0, 2.0)	-	-	25.5	-
Group 2 - Sum	-	375.0	-	495.5	PASSED
Zone III	-	-	625.0	594.4	PASSED
Zone IV	-	1700.0	-	2373.6	PASSED
Zone I	-	-	17600.0	17507.7	PASSED

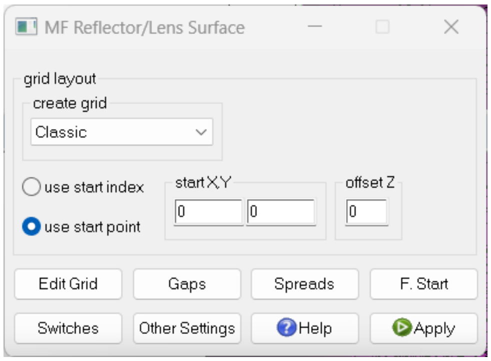
Table S4. Test report for ECE R112 class-B driving-beam distribution

Test Item	(u, v)	Minimum(cd)	Maximum(cd)	Actual(cd)	Status
Point B 50 L	(-3.4, 0.6)	-	350.0	104.6	PASSED
Point BR	(-2.5, 1.0)	-	1750.0	95.9	PASSED
Point 75 R	(1.1, -0.6)	10100.0	-	12104.0	PASSED
Point 75 L	(-3.4, -0.6)	-	10600.0	1418.4	PASSED
Point 50 L	(-3.4, -0.9)	-	13200.0	2843.2	PASSED
Point 50 R	(1.7, -0.9)	-	10100.0	6592.9	PASSED
Point 50 V	(0.0, -0.9)	5100.0	-	12606.4	PASSED
Point 25 L	(-9.0, -1.7)	1700.0	-	3974.9	PASSED
Point 25 R	(9.0, -1.7)	1700.0	-	4655.3	PASSED
Point 7	(-8.0, 0.0)	65.0	-	209.1	PASSED
Point 8	(-4.0, 0.0)	125.0	-	179.5	PASSED
Group 1 - Point 1	(-8.0, 4.0)	-	-	461.4	-
Group 1 - Point 2	(0.0, 4.0)	-	-	388.9	-
Group 1 - Point 3	(8.0, 4.0)	-	-	322.5	-
Group 1 - Sum	-	190.0	-	1172.7	PASSED
Group 2 - Point 4	(-4.0, 2.0)	-	-	235.6	-
Group 2 - Point 5	(0.0, 2.0)	-	-	210.1	-
Group 2 - Point 6	(4.0, 2.0)	-	-	221.0	-
Group 2 - Sum	-	375.0	-	666.7	PASSED
Zone III	-	-	625.0	625.0	PASSED
Zone IV	-	2500.0	-	2503.4	PASSED
Zone I	-	-	13185.7	11740.6	PASSED

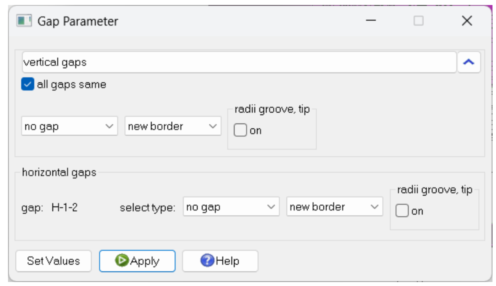
2. EXAMPLE REFLECTOR DESIGN AND SIMULATION IN LIGHTTOOLS

To demonstrate the capability of our system to model, simulate, and optimize freeform surface arrays, we conducted comparative experiments using the *Macro Focal Reflector* feature within the *Advanced Design Features* of *LightTools 2022.03*. In Figure S1, we present the configuration dialog of the MF reflector, which mainly includes the base grid settings, spread settings, and gap settings. The base grid size of the entire reflector is 9 mm in width and 6 mm in height, with 3 facets in the u direction and 2 facets in the v direction, all set to *no gap*. We used a point light source positioned at [0, 0, 2] and rotated 180° around the x-axis as the design light source for the MF reflector. The facet spread parameters were set to [-50, 50]° in the horizontal direction and [-20, 20]° in the vertical direction. The farfield intensity receiver range was set to $[-60°, 60°]$ and $[-30°, 30°]$,with a resolution of 240×120. As shown in Fig. S2, we used the simulated light distribution of this reflector as the optimization target for differentiable ray tracing to design the example reflector.

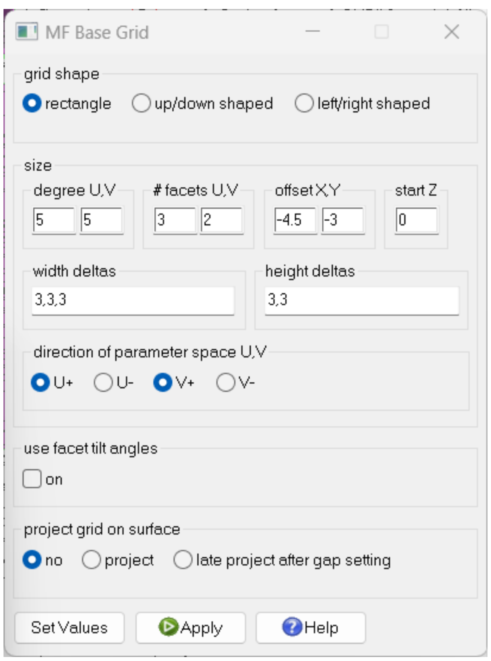
We present the design and simulation result of the example reflector in Fig. S3. The reflector model designed through automated differential optimization is similar to the reference model, and the distribution and details of the reflected light match the range and pattern of the reference light distribution.



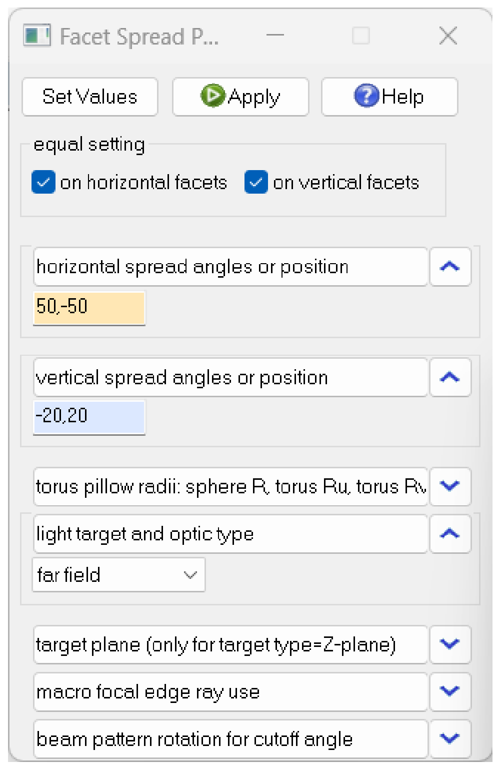
(a) MF reflector entrance setting dialog.



(b) MF reflector gap setting dialog.

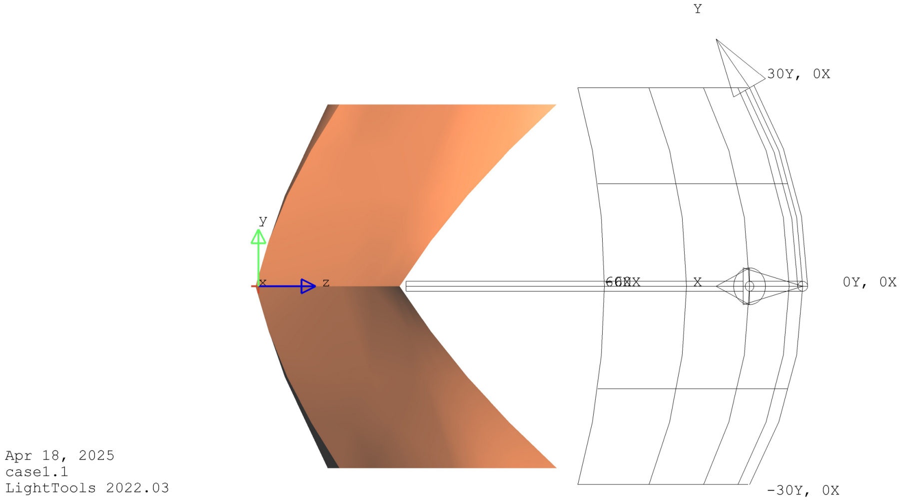


(c) MF reflector base grid setting dialog.

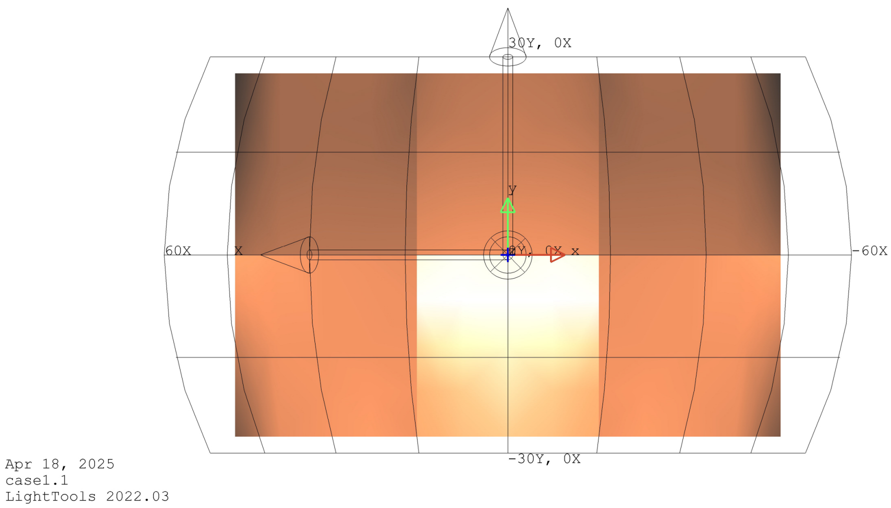


(d) MF reflector spread setting dialog.

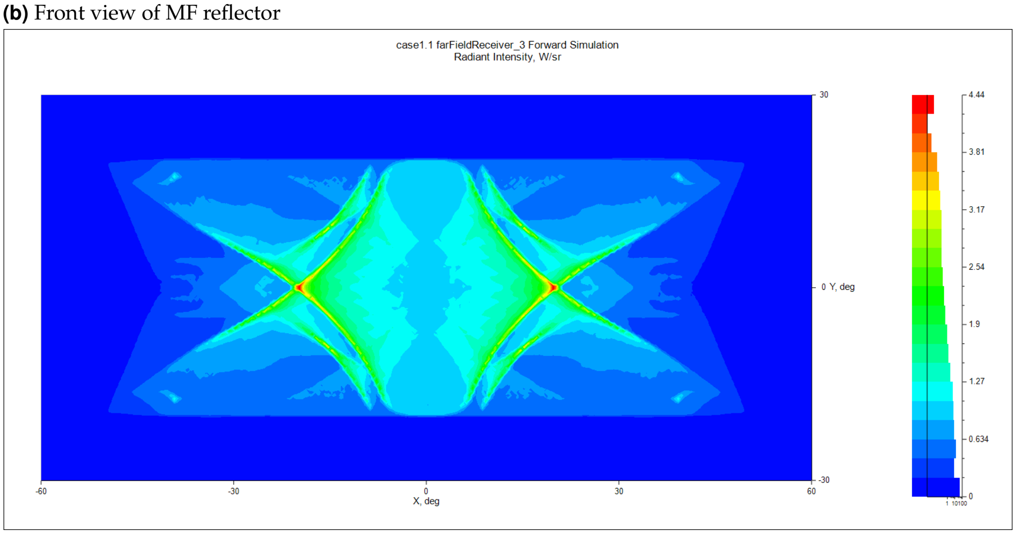
Fig. S1. MF reflector functions in *LightTools*.



(a) Side view of MF reflector

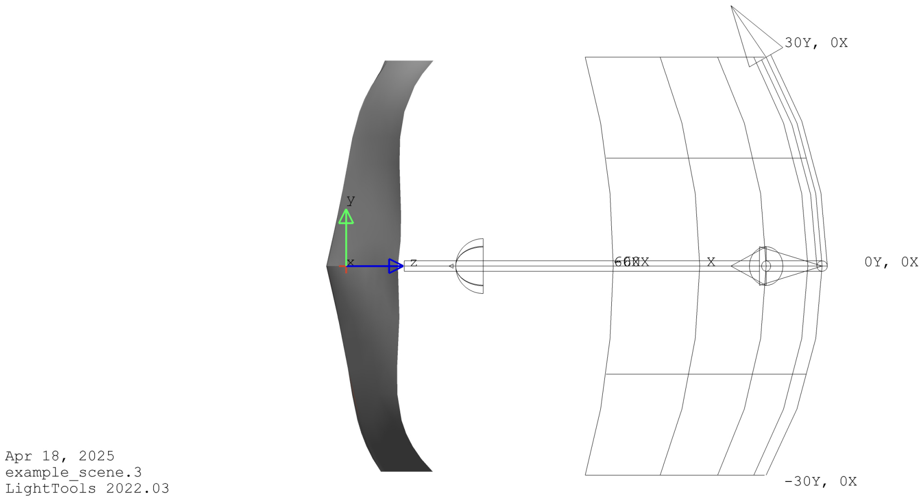


(b) Front view of MF reflector

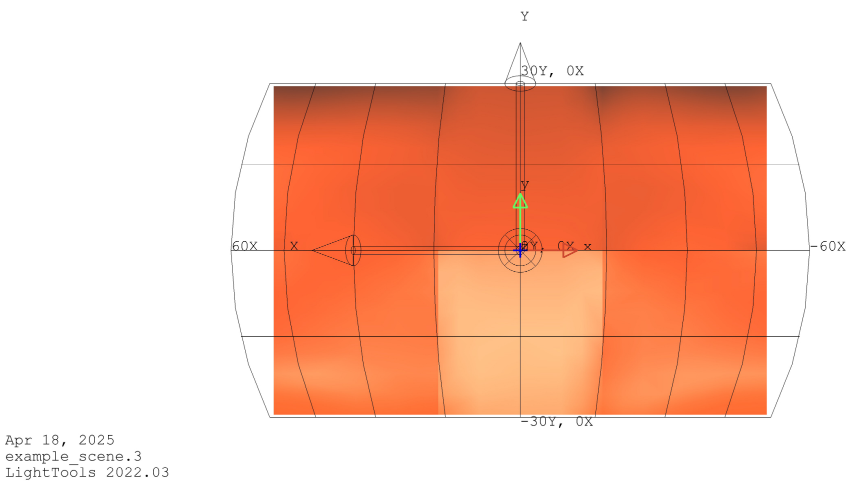


(c) Intensity Sensor of MF reflector

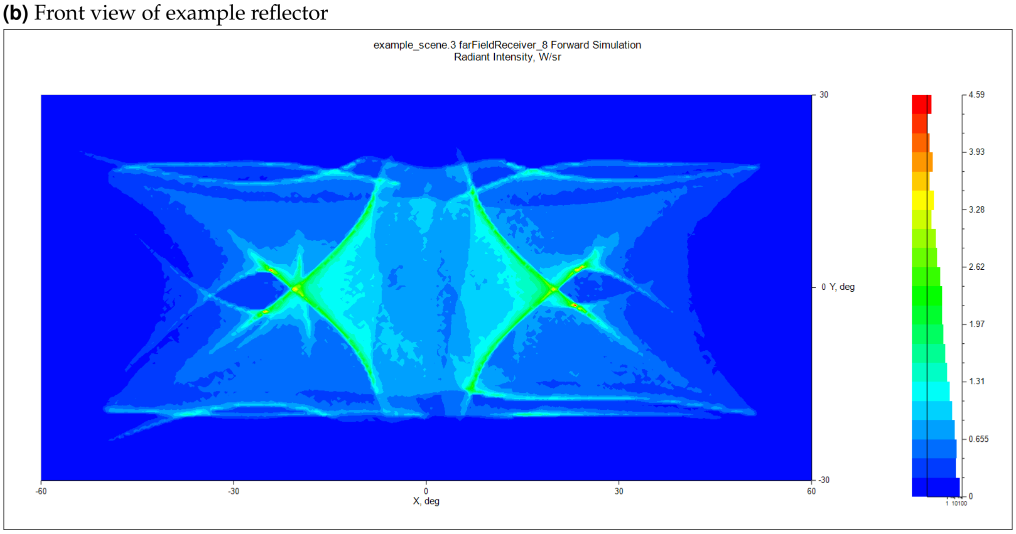
Fig. S2. MF reflector modeling and simulation result.



(a) Side view of example reflector



(b) Front view of example reflector

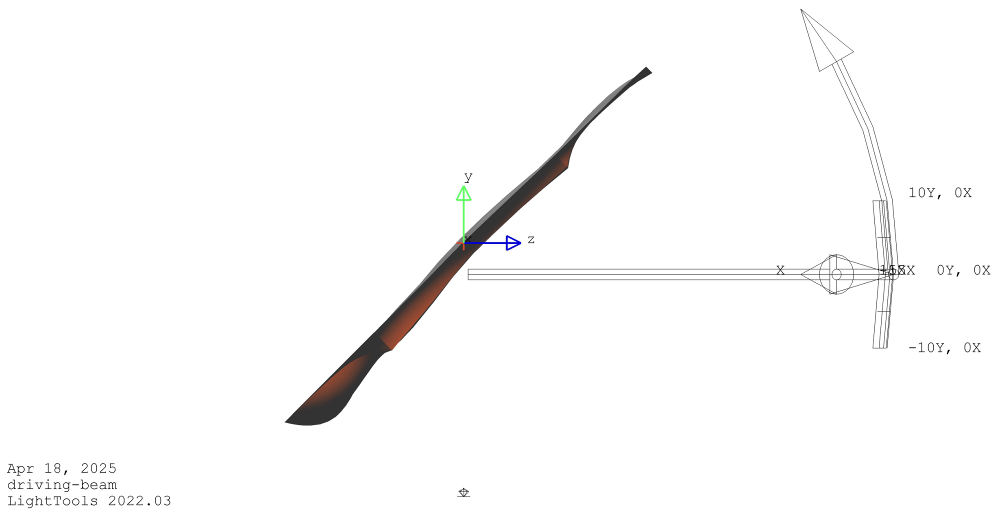


(c) Intensity Sensor of example reflector

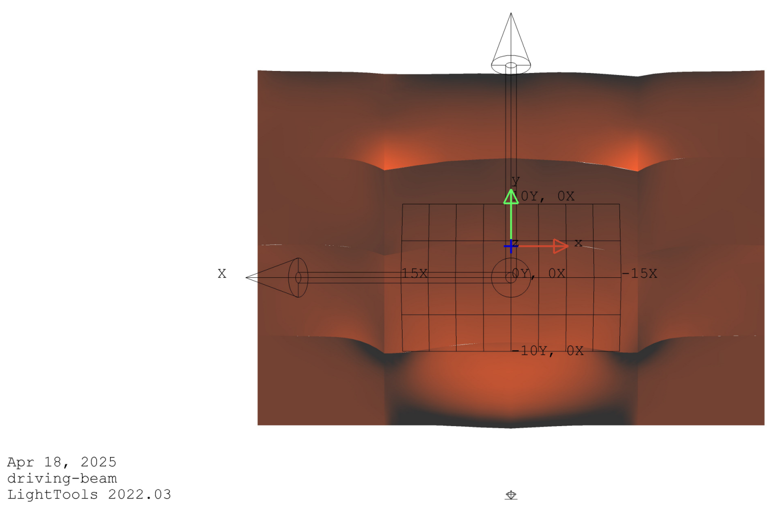
Fig. S3. Example reflector modeling and simulation result.

3. DRIVING-BEAM REFLECTOR DESIGN AND SIMULATION IN LIGHTTOOLS

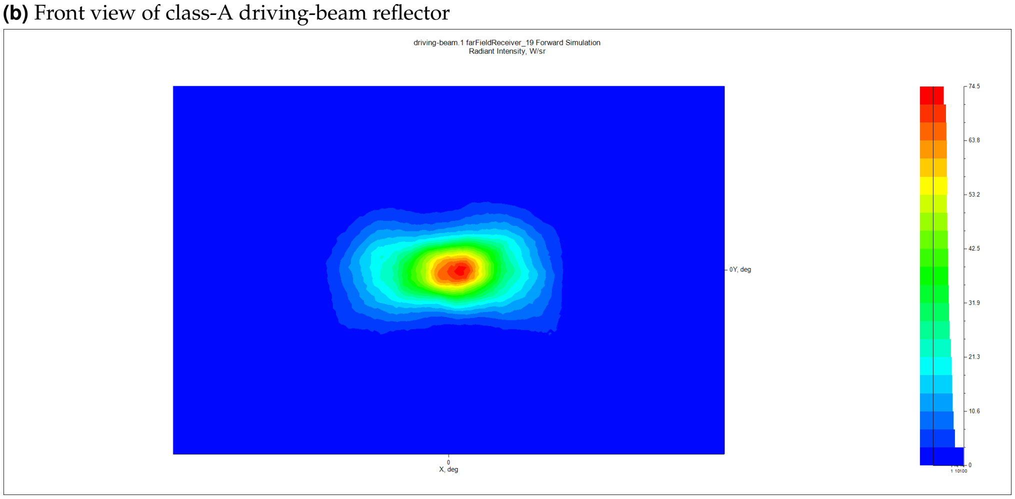
We present the CAD models in STEP format and the simulation results in LightTools for our designed driving-beam reflector used in class-A and class-B headlamps, as shown in Figs. S4 and S5.



(a) Side view of class-A driving-beam reflector

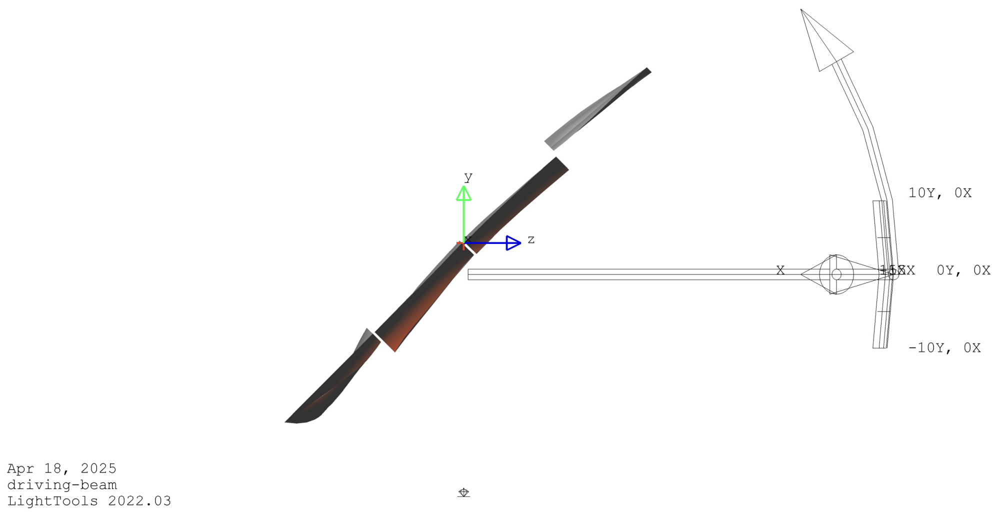


(b) Front view of class-A driving-beam reflector

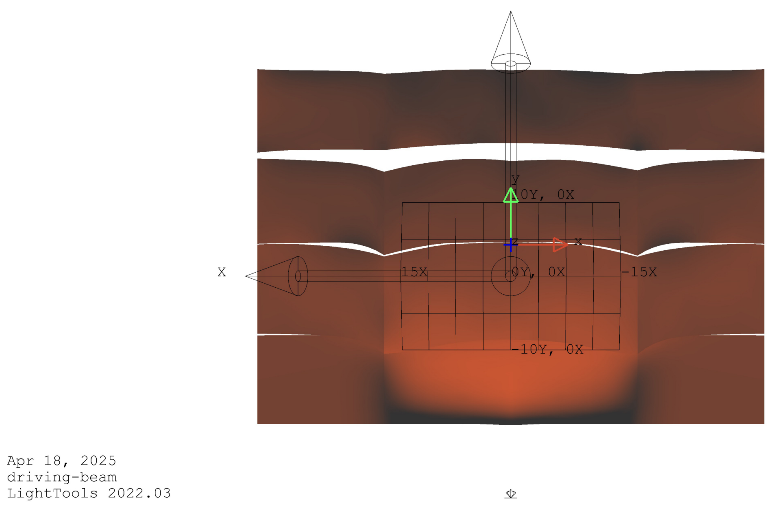


(c) Intensity Sensor of class-A driving-beam reflector

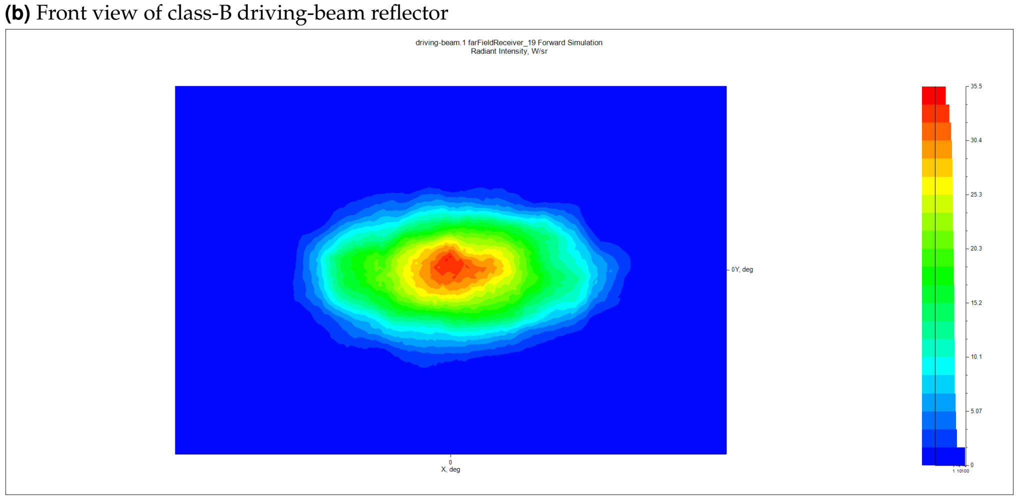
Fig. S4. Class-A driving-beam reflector modeling and simulation result.



(a) Side view of class-B driving-beam reflector



(b) Front view of class-B driving-beam reflector

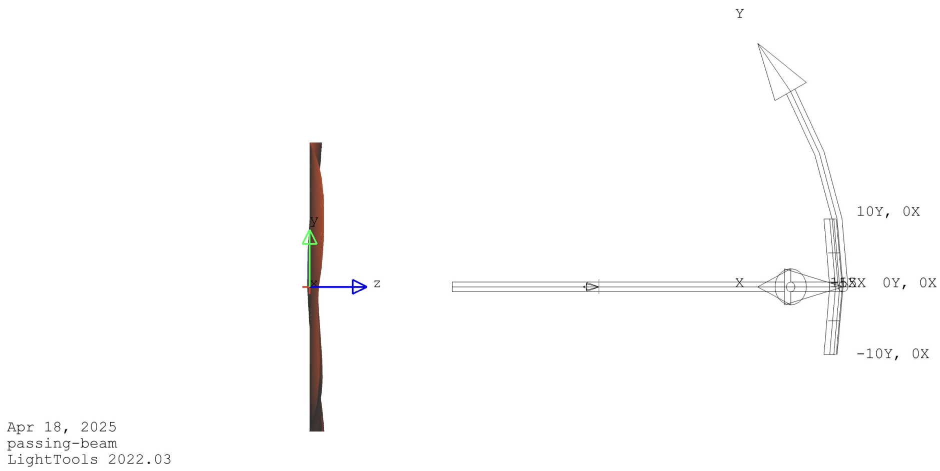


(c) Intensity Sensor of class-B driving-beam reflector

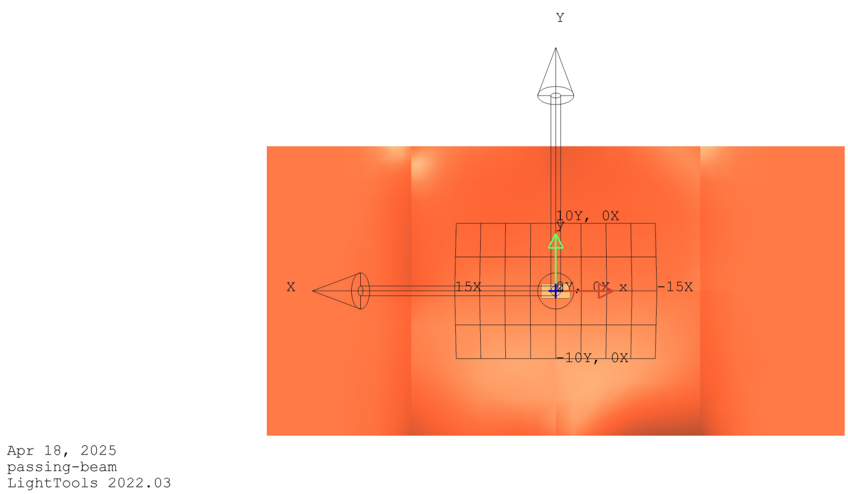
Fig. S5. Class-B driving-beam reflector modeling and simulation result.

4. PASSING-BEAM REFLECTOR DESIGN AND SIMULATION IN LIGHTTOOLS

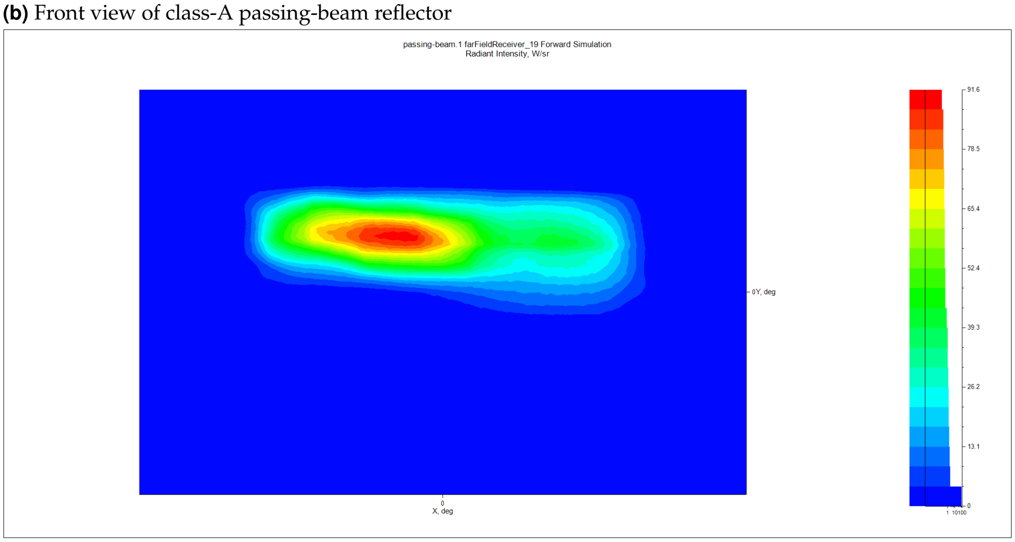
We present the CAD models in STEP format and the simulation results in LightTools for our designed passing-beam reflector used in class-A and class-B headlamps, as shown in Figs. S6 and S7.



(a) Side view of class-A passing-beam reflector

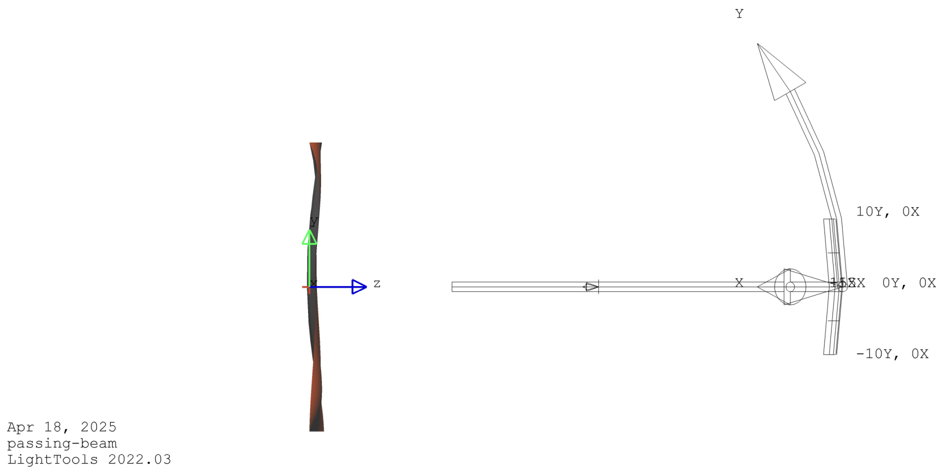


(b) Front view of class-A passing-beam reflector

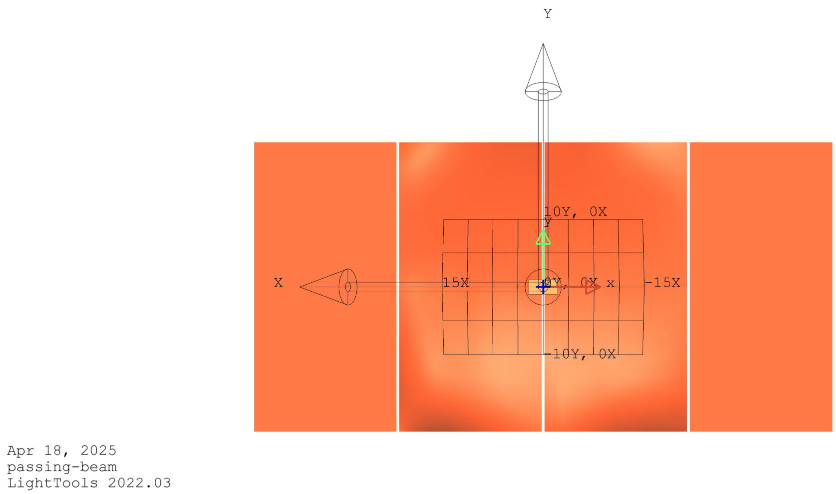


(c) Intensity Sensor of class-A passing-beam reflector

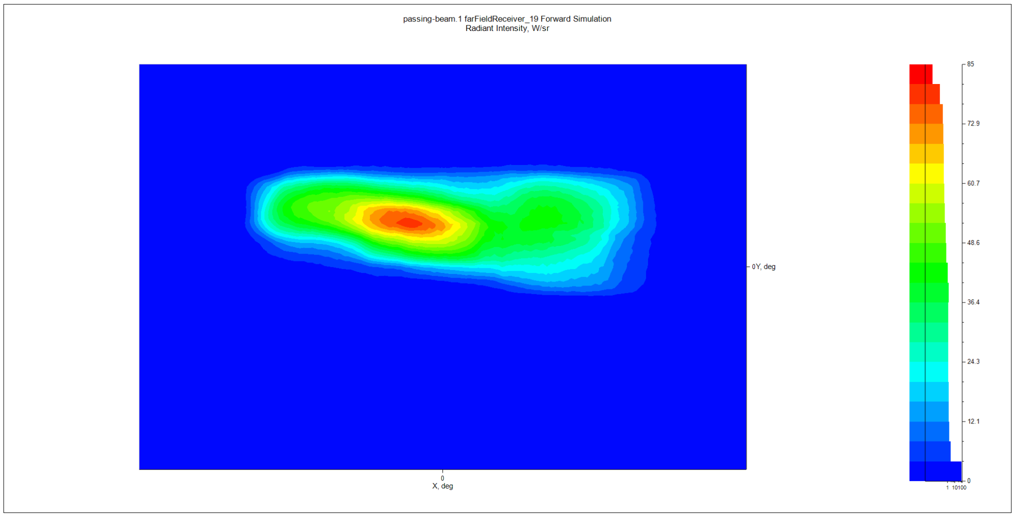
Fig. S6. Class-A passing-beam reflector modeling and simulation result.



(a) Side view of class-B passing-beam reflector



(b) Front view of class-B passing-beam reflector



(c) Intensity Sensor of class-B passing-beam reflector

Fig. S7. Class-B passing-beam reflector modeling and simulation result.

5. REGULATION TEST REPORTS OF DESIGNED REFLECTORS

Table S5. Test report for ECE R112 class-A passing-beam reflector

Test Item	(u, v)	Minimum(cd)	Maximum(cd)	Actual(cd)	Status
Point B 50 L	(-3.4, 0.6)	-	350.0	153.6	PASSED
Point BR	(-2.5, 1.0)	-	1750.0	153.7	PASSED
Point 75 R	(1.1, -0.6)	5100.0	-	4154.3	FAILED
Point 75 L	(-3.4, -0.6)	-	10600.0	1660.5	PASSED
Point 50 L	(-3.4, -0.9)	-	13200.0	3658.1	PASSED
Point 50 R	(1.7, -0.9)	-	5100.0	4662.8	PASSED
Point 50 V	(0.0, -0.9)	5100.0	-	5661.5	PASSED
Point 25 L	(-9.0, -1.7)	1250.0	-	2672.1	PASSED
Point 25 R	(9.0, -1.7)	1250.0	-	2630.8	PASSED
Point 7	(-8.0, 0.0)	65.0	-	573.2	PASSED
Point 8	(-4.0, 0.0)	125.0	-	212.7	PASSED
Group 1 - Point 1	(-8.0, 4.0)	-	-	23.7	-
Group 1 - Point 2	(0.0, 4.0)	-	-	47.4	-
Group 1 - Point 3	(8.0, 4.0)	-	-	11.8	-
Group 1 - Sum	-	190.0	-	82.9	FAILED
Group 2 - Point 4	(-4.0, 2.0)	-	-	94.6	-
Group 2 - Point 5	(0.0, 2.0)	-	-	189.2	-
Group 2 - Point 6	(4.0, 2.0)	-	-	147.8	-
Group 2 - Sum	-	375.0	-	431.6	PASSED
Zone III	-	-	625.0	1465.4	FAILED
Zone IV	-	1700.0	-	2446.7	PASSED
Zone I	-	-	17600.0	16752.4	PASSED

Table S6. Test report for ECE R112 class-B driving-beam reflector

Test Item	(u, v)	Minimum(cd)	Maximum(cd)	Actual(cd)	Status
Point B 50 L	(-3.4, 0.6)	-	350.0	330.9	PASSED
Point BR	(-2.5, 1.0)	-	1750.0	271.9	PASSED
Point 75 R	(1.1, -0.6)	10100.0	-	7032.1	FAILED
Point 75 L	(-3.4, -0.6)	-	10600.0	2440.6	PASSED
Point 50 L	(-3.4, -0.9)	-	13200.0	4426.4	PASSED
Point 50 R	(1.7, -0.9)	-	10100.0	7594.0	PASSED
Point 50 V	(0.0, -0.9)	5100.0	-	8510.0	PASSED
Point 25 L	(-9.0, -1.7)	1700.0	-	2577.5	PASSED
Point 25 R	(9.0, -1.7)	1700.0	-	2955.9	PASSED
Point 7	(-8.0, 0.0)	65.0	-	224.5	PASSED
Point 8	(-4.0, 0.0)	125.0	-	502.3	PASSED
Group 1 - Point 1	(-8.0, 4.0)	-	-	47.4	-
Group 1 - Point 2	(0.0, 4.0)	-	-	100.7	-
Group 1 - Point 3	(8.0, 4.0)	-	-	165.9	-
Group 1 - Sum	-	190.0	-	314.0	PASSED
Group 2 - Point 4	(-4.0, 2.0)	-	-	319.3	-
Group 2 - Point 5	(0.0, 2.0)	-	-	153.7	-
Group 2 - Point 6	(4.0, 2.0)	-	-	609.0	-
Group 2 - Sum	-	375.0	-	1082.0	FAILED
Zone III	-	-	625.0	3829.1	PASSED
Zone IV	-	2500.0	-	2931.3	PASSED
Zone I	-	-	13185.7	14079.4	PASSED

Table S7. Test report for ECE R112 class-A driving-beam reflector

Test Item	(u, v)	Minimum(cd)	Maximum(cd)	Actual(cd)	Status
Point H-5L	(-5.0, 0.0)	3400.0	-	6598.4	PASSED
Point H-2,5L	(-2.5, 0.0)	13500.0	-	16988.5	PASSED
Point H-2,5R	(2.5, 0.0)	13500.0	-	18055.4	PASSED
Point H-5R	(5.0, 0.0)	3400.0	-	5071.9	PASSED
Imax	(0.1, -0.0)	27000	215000	37046.5	PASSED

Table S8. **Test report for ECE R112 class-B driving-beam reflector**

Test Item	(u, v)	Minimum(cd)	Maximum(cd)	Actual(cd)	Status
Point H-5L	(-5.0, 0.0)	5100.0	-	19532.7	PASSED
Point H-2,5L	(-2.5, 0.0)	20300.0	-	28658.9	PASSED
Point H-2,5R	(2.5, 0.0)	20300.0	-	28297.8	PASSED
Point H-5R	(5.0, 0.0)	5100.0	-	15560.5	PASSED
Imax	(0.1, -0.0)	40500	215000	37326.2	PASSED

6. AN EXPERIMENTAL CASE FOR B-SPLINE SURFACE TRIANGULATION DEVIATION

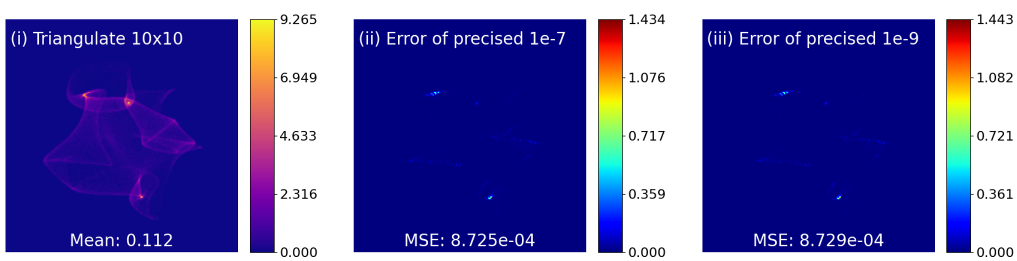
To illustrate the deviation between the outgoing rays from precise B-Spline simulation and our triangulated simulation, we conducted an additional experiment using a single surface reflector. The size of this reflector is $10mm \times 10mm$ with 10 control points in the u- and v-direction. The degree of the B-Spline basis is 3 for both directions. We used a point light source positioned at $[0, 0, 5]$ and rotated 180° around the x-axis, emitting rays over the upper hemisphere range in $[0°, 40°]$. The farfield intensity receiver range was set to $[-90°, 90°]$ and $[-90°, 90°]$, with a resolution of 180×180.

In this case, we conducted an iterative numerical calculation for precise ray-surface intersection using two different error thresholds, specifically 1e-7 and 1e-9. Given the tensor-based nature of our simulation, the rays intersect in parallel. Therefore, the thresholds refer to the average distance between the intersection points of the rays and the points on the B-spline surface. Once the average error of the intersection points for a ray falls below this threshold, it is considered that the precise solution has been found. We compared the errors in the origin points and directions of the outgoing rays from the triangulated intersections and the precise intersections with different thresholds. We used the average L2 norm as an error metric, and the detailed results are shown in Table S9.

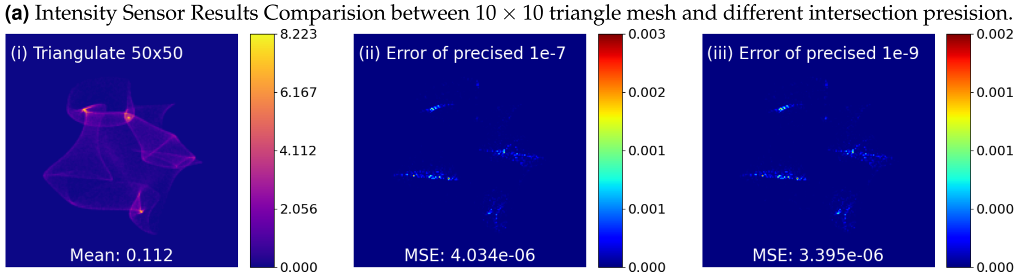
In this case, we present the simulation results of the intensity sensor grids and visually compare the differences in accuracy between triangulated intersections and precise intersections for intensity sensors. As shown in Fig. S8(a), we present the error map between the 10×10 triangulated intensity sensor simulation results and the precise intersections. We have calculated the Mean value and the Mean Squared Error (MSE), where, at this triangulation resolution, the average error with precise intersections is 1e-4. In Fig. S8(b), we present the results for the 50×50 triangle mesh, with error ranges in the order of 1e-6. The results presented above indicate that after statistically analyzing the direction of light rays using intensity sensor grids, the quantifiable error margins were further reduced. This demonstrates the feasibility of using refined triangulated meshes for intersection calculations in terms of both performance and accuracy. We also present the model and simulation result of the experimental B-Spline reflector in LightTools Fig. S9.

Triangulation Resolution	Num rays	Threshold	Origin deviation	Direction deviation
10×10	1e5	1e-7	9.22076e-3	3.72661e-3
10×10	1e5	1e-9	9.22083e-3	3.72617e-3
10×10	1e6	1e-7	9.19060e-3	3.61975e-3
10×10	1e6	1e-9	9.18707e-3	3.61914e-3
50×50	1e5	1e-7	4.43643e-4	1.70489e-4
50×50	1e5	1e-9	4.22330e-4	1.61409e-4
50×50	1e6	1e-7	4.14105e-4	1.62562e-4
50×50	1e6	1e-9	4.18625e-4	1.59506e-4

Table S9. Deviation of outgoing rays origins and directions between triangulate intersection and precised intersection

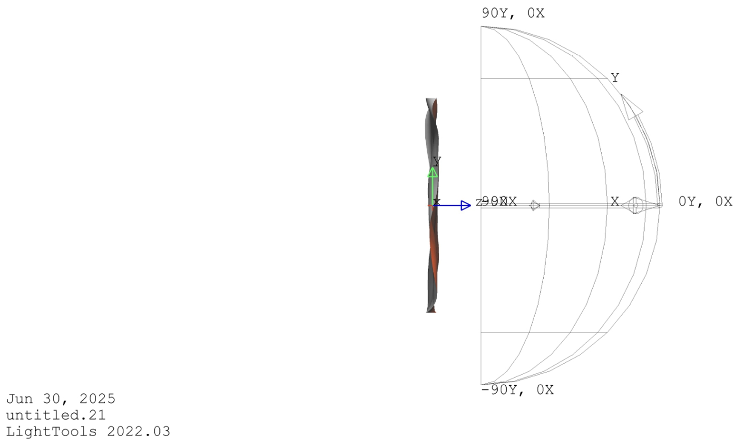


(a) Intensity Sensor Results Comparision between 10×10 triangle mesh and different intersection presision.

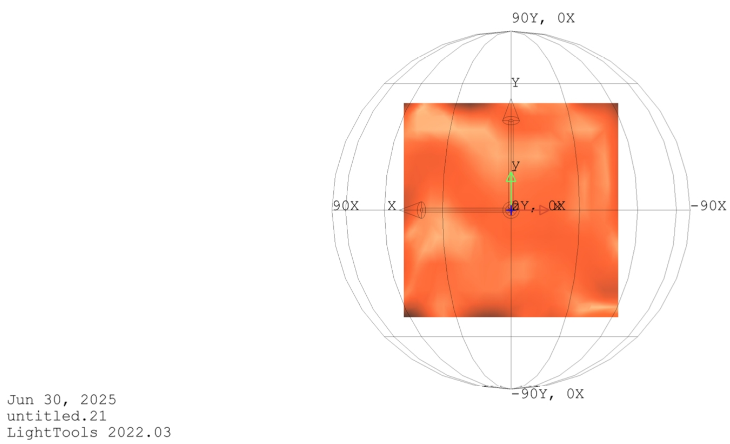


(b) Intensity Sensor Results Comparision between 50×50 triangle mesh and different intersection presision.

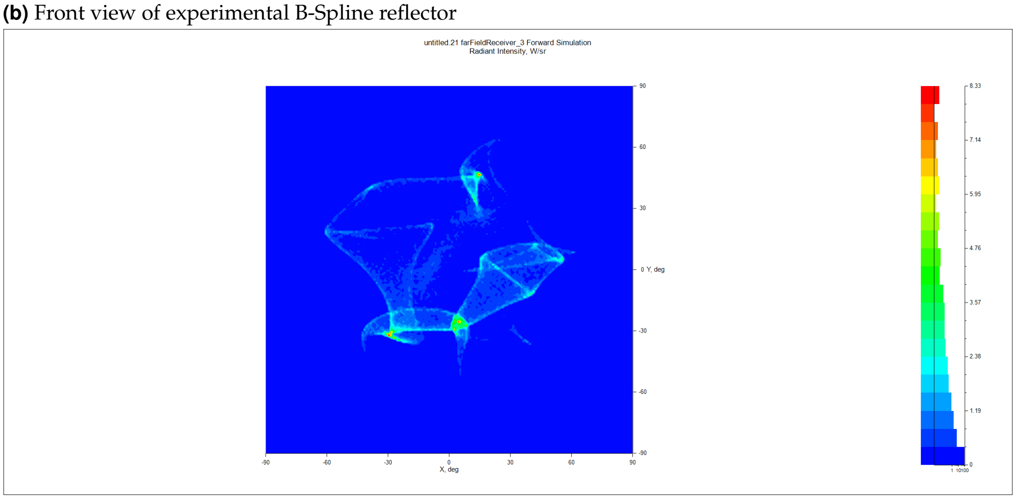
Fig. S8. The error of intentisy sensor between triangle mesh and presice simulation



(a) Side view of experimental B-Spline reflector



(b) Front view of experimental B-Spline reflector



(c) Intensity Sensor of experimental B-Spline reflector

Fig. S9. Experimental B-Spline reflector modeling and simulation result.